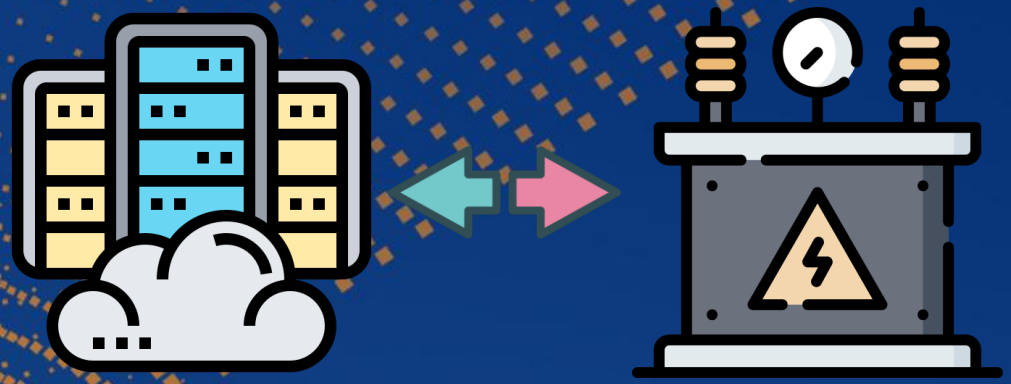
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# AI Data Centers Need More Than Power:

## Transformers and Power Electronics Are Hidden Supply Chain Externalities

**Boyu Yao**
Joint work with Prof. Yury Dvorkin

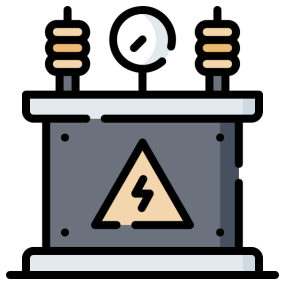
**November 2026**

**Department of Civil and Systems Engineering, Johns Hopkins University**
**Ralph O'Connor Sustainable Energy Institute (ROSEI)**
**Critical Minerals Research Lab, Resources for the Future (RFF)**
**Electric Power Innovation for a Carbon-free Society (EPICS)**
**Academic Center for Reliability and Resilience of Offshore Wind (ARROW)**
**Smart Energy reseARCH (SEARCH) Group**

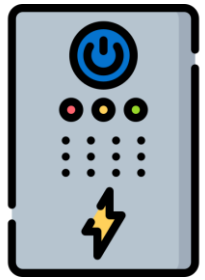
Energy at Hopkins

# Different projects can still compete upstream.

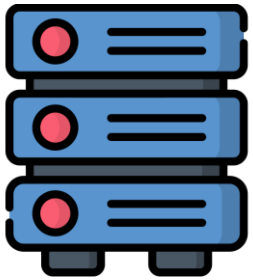
*These illustrations were generated using ChatGPT under the authors' direction.



Transformer



UPS



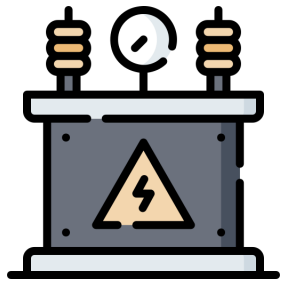
Servers



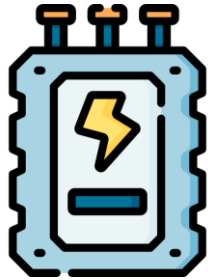
A new AI data center



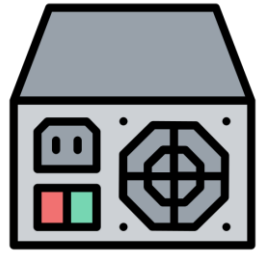
An existing utility substation



Transformer



Converter/Inverter



PCS

**Shared materials, manufacturing, and delivery capacity**

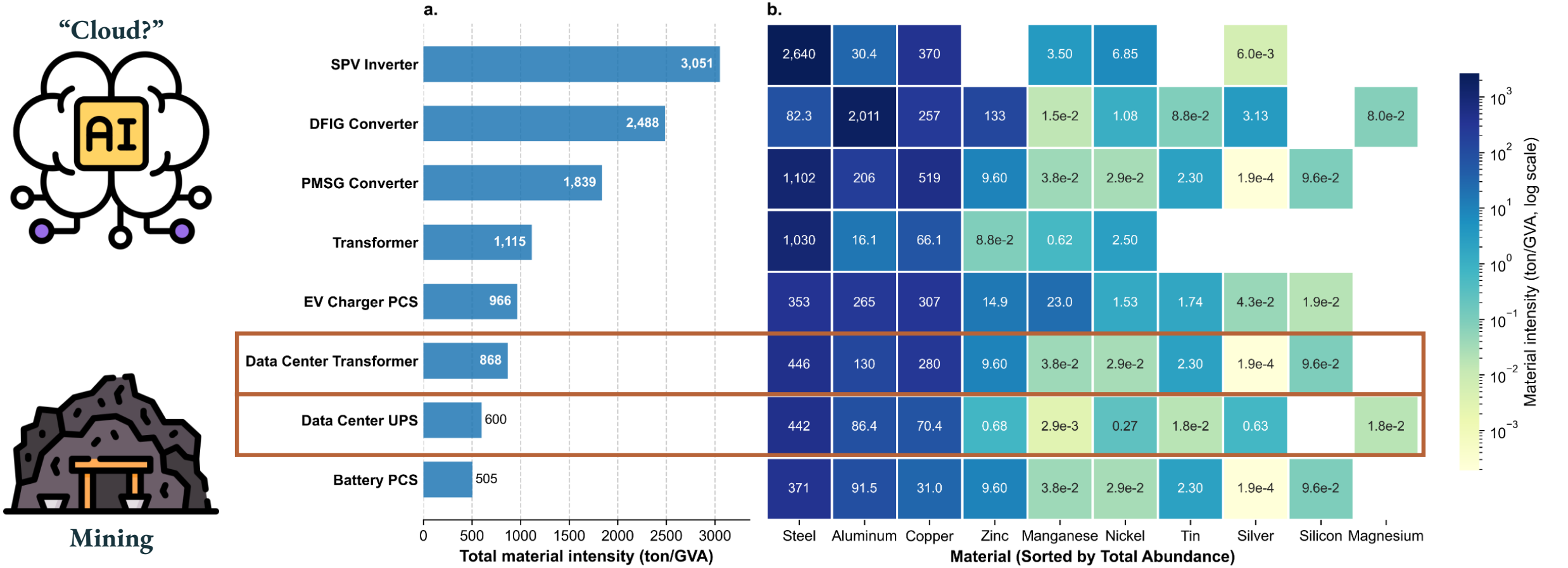
Copper & steel · transformer / UPS production · lead times and replacement schedules

**Average power transformer lead times: ~1 year (2021) to ~2.3 years (2024)**
**Large units can face 3–4-year waits, as reported by DOE in 2026**

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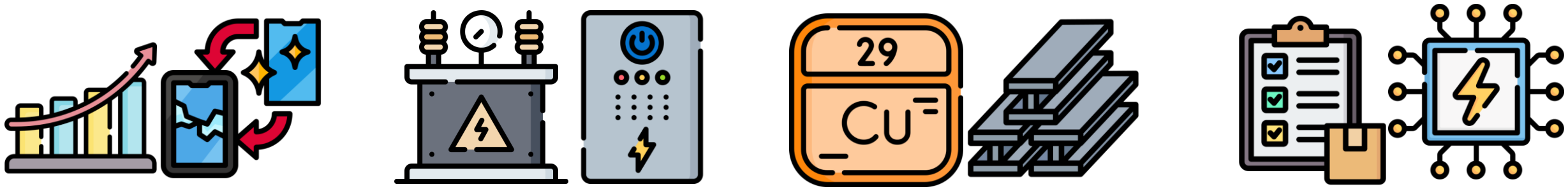
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# Data centers need critical materials, and so does the rest of the grid.




Material requirements per unit of equipment capacity (tonnes/GVA)

# What has to happen before a megawatt becomes usable?




**New build + replacement** → **Equipment demand** → **Materials required** → **Deliverable capacity**

Track the equipment fleet

Grid-supporting equipment (GSE)

Apply equipment material intensities

Respect accessible material supply

## Can the supply chain support everything we plan to build?

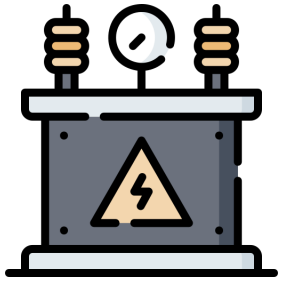
U.S. case study through 2030 · MRSUT electronics-sector allocations used to estimate material availability for GSE.

WHERE THE PRESSURE APPEARS FIRST

# Data center equipment faces early shortages.

2030 · U.S. HIGH GROWTH · OPTIMISTIC LIFETIMES

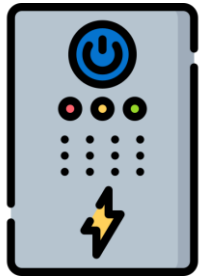
Unmet required GSE capacity



**Data center transformers**

**76.4%**

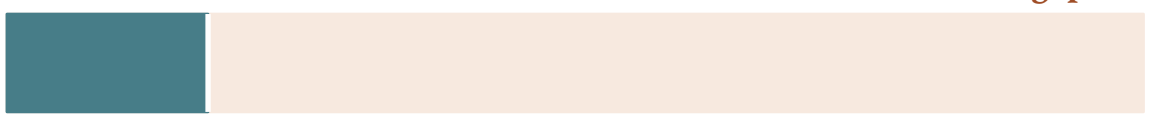
**14.1 GVA gap**



**Data-center UPS**

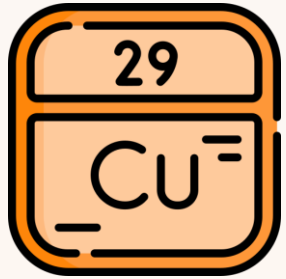
**82.3%**

**22.1 GVA gap**



Supplied Unmet

**FIRST BINDING MATERIAL**




**COPPER**

**100%**

of modeled GSE-accessible supply used by 2027

**By 2030: Steel, 96.7%; Nickel, 93.0%; Manganese, 73.1%.**

**Why does load-side equipment face shortages first?**

When materials are scarce, the model prioritizes bulk transformers, inverters, and converters over data center transformers and UPS systems.

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# By 2030, the same constraint reaches the broader grid.

2030 · U.S. HIGH GROWTH · OPTIMISTIC LIFETIMES

## 28.5%

**of modeled equipment demand is unmet**




269.6 GVA gap · 946.6 GVA required · 677.0 GVA deliverable
107.3 GVA bulk transformers · 63.6 GVA inverters · 57.1 GVA converters
21.5 GVA DC UPS · 10.5 GVA DC transformers · 9.7 GVA EV charger PCS

Shortages widen over time:

**2027: Data-center equipment + wind converters**
**2030: Shortages reach bulk transformers**

**Even priority bulk transformers are affected**

0 → **107.3**

2027

GVA gap in 2030

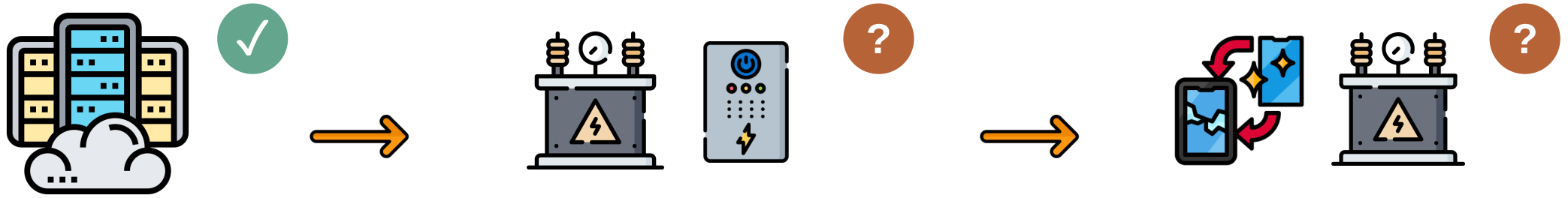
14.5% of bulk transformer need

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# Data center growth is a grid-wide supply chain issue.

Serving new data centers draws on the same equipment and materials needed elsewhere on the grid.



**Power available**

generation + interconnection

**Equipment deliverable?**

transformer + UPS + power electronics

**copper · steel · factories · lead time**

**Other grid needs still met?**

replacement + expansion

**A plan can be affordable and electrically feasible, and still not be buildable on time.**

**Yao, B., & Dvorkin, Y. (2026). Grid-Supporting Equipment Supply Chains Constrain the Feasible Pace of Power System Expansion.**

***Cell Reports Sustainability (accepted, in press)***

***Preprint: arXiv:2604.18411***